KAIZEN MATHS

Lesson starter

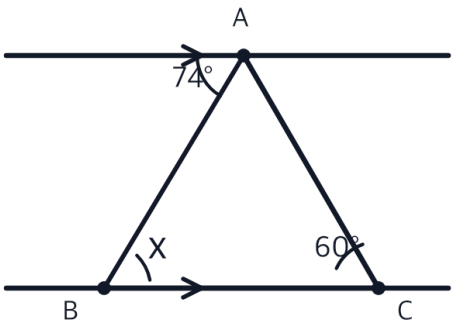
Formal Geometric Proof

Formal Proofs

Name _____

SA1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 46°.

GEO-EB04-L4 · SA

Show your working

KAIZEN MATHS

Lesson starter

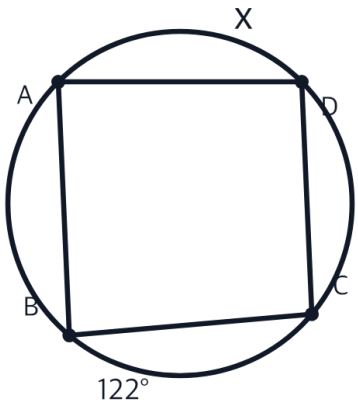
Formal Geometric Proof

Formal Proofs

Name _____

SB1

Complete a two-column proof using circle-theorem reasons.



ABCD is cyclic. Complete the proof that x = 58°.

KAIZEN MATHS

Lesson starter

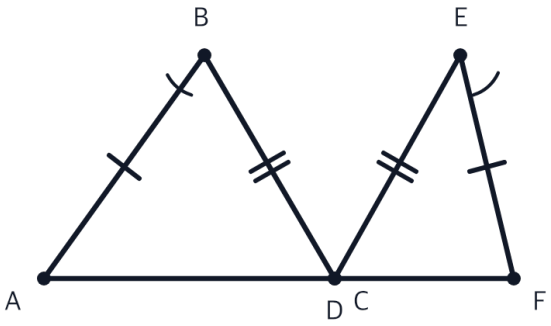
Formal Geometric Proof

Formal Proofs

Name _____

SC1

Write the proof as a connected paragraph.



Write a paragraph proof that triangle ABC is congruent to triangle EFD.

GEO-EB04-L4 · SC

Show your working

KAIZEN MATHS

Lesson starter

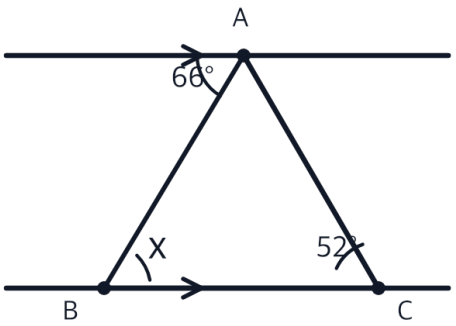
Formal Geometric Proof

Formal Proofs

Name _____

SD1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 62°.

GEO-EB04-L4 · SD

Show your working

KAIZEN MATHS

Exit ticket

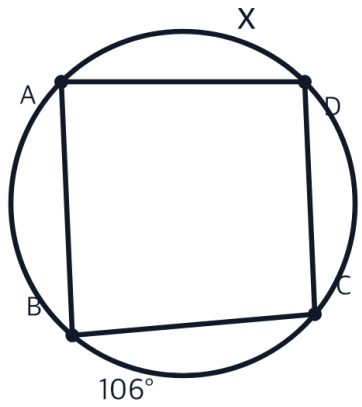
Formal Geometric Proof

Formal Proofs

Name _____

EA1

Complete a two-column proof using circle-theorem reasons.



ABCD is cyclic. Complete the proof that x = 74°.

Show your working

KAIZEN MATHS

Exit ticket

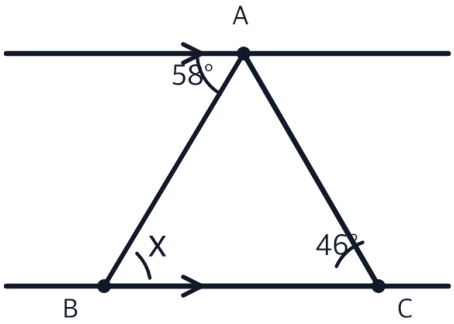
Formal Geometric Proof

Formal Proofs

Name _____

EB1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 76°.

Show your working

KAIZEN MATHS

Exit ticket

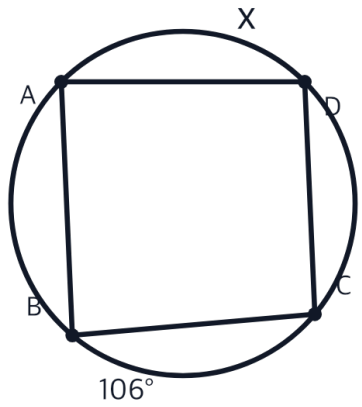
Formal Geometric Proof

Formal Proofs

Name _____

EA1

Complete a two-column proof using circle-theorem reasons.



ABCD is cyclic. Complete the proof that x = 74°.

GEO-EB04-L4 · EA

Show your working

KAIZEN MATHS

Exit ticket

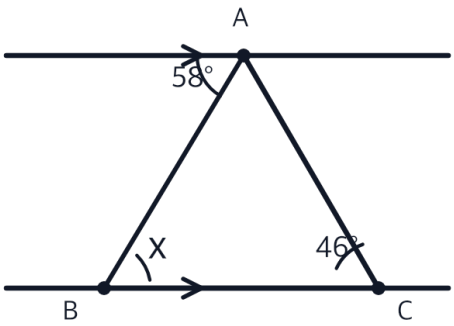
Formal Geometric Proof

Formal Proofs

Name _____

EB1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 76°.

GEO-EB04-L4 · EB

Show your working

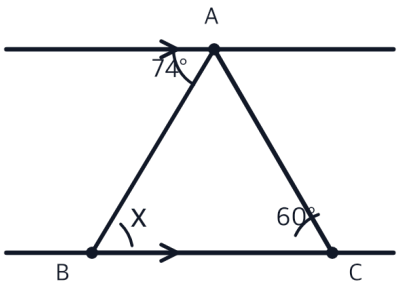
Teacher answers & working

Formal Geometric Proof · Formal Proofs

GEO-EB04-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 46°.

Answer:

46°

Working:

Statement	Reason
The two marked angles on the parallel lines are equal.	Alternate interior angles are equal.
One angle in the triangle is 74°.	Substitute the corresponding angle value.
The second given angle is 60°.	Given.
$x + 74° + 60° = 180°$	Angles in a triangle add to 180°.
$x = 46°$	Solve the equation.

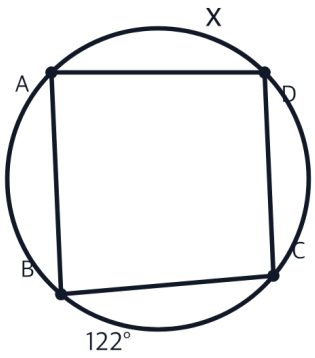
Teacher answers & working

Formal Geometric Proof · Formal Proofs

GEO-EB04-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1

Complete a two-column proof using circle-theorem reasons.



ABCD is cyclic. Complete the proof that x = 58°.

Answer:

58°

Working:

Statement	Reason
ABCD is cyclic.	Given by the diagram.
Angle B + angle D = 180°	Opposite angles in a cyclic quadrilateral are supplementary.
$122° + x = 180°$	Substitute angle B = 122°.
$x = 180° - 122°$	Subtract 122° from both sides.
$x = 58°$	Calculate.

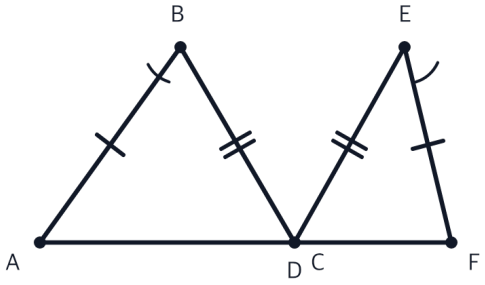
Teacher answers & working

Formal Geometric Proof · Formal Proofs

GEO-EB04-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1

Write the proof as a connected paragraph.



Write a paragraph proof that triangle ABC is congruent to triangle EFD.

Answer:

Triangle ABC is congruent to triangle EFD by SAS.

Working:

From the diagram, AB = EF and BC = FD.

The included angles are also equal: angle ABC = angle EFD.

The equal angle is between the two equal sides in each triangle.

Therefore, by side-angle-side congruence, triangle ABC is congruent to triangle EFD.

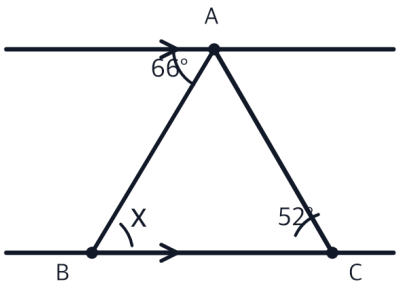
Teacher answers & working

Formal Geometric Proof · Formal Proofs

GEO-EB04-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 62°.

Answer:

62°

Working:

Statement	Reason
The two marked angles on the parallel lines are equal.	Alternate interior angles are equal.
One angle in the triangle is 66°.	Substitute the corresponding angle value.
The second given angle is 52°.	Given.
$x + 66° + 52° = 180°$	Angles in a triangle add to 180°.
$x = 62°$	Solve the equation.

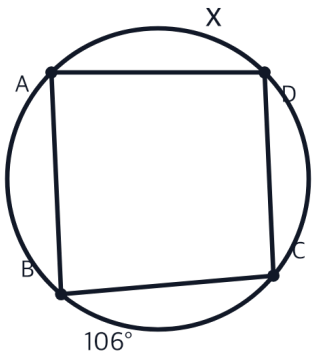
Teacher answers & working

Formal Geometric Proof · Formal Proofs

GEO-EB04-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1

Complete a two-column proof using circle-theorem reasons.



ABCD is cyclic. Complete the proof that x = 74°.

Answer:

74°

Working:

Statement	Reason
ABCD is cyclic.	Given by the diagram.
Angle B + angle D = 180°	Opposite angles in a cyclic quadrilateral are supplementary.
$106° + x = 180°$	Substitute angle B = 106°.
$x = 180° - 106°$	Subtract 106° from both sides.
$x = 74°$	Calculate.

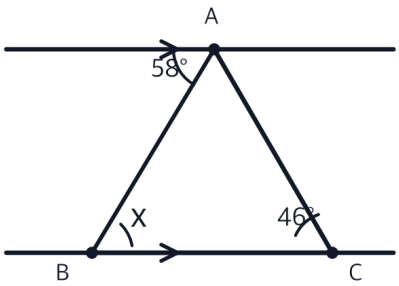
Teacher answers & working

Formal Geometric Proof · Formal Proofs

GEO-EB04-L4 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1

Complete a two-column proof with a statement and reason in every row.



Complete the proof that x = 76°.

Answer:

76°

Working:

Statement	Reason
The two marked angles on the parallel lines are equal.	Alternate interior angles are equal.
One angle in the triangle is 58°.	Substitute the corresponding angle value.
The second given angle is 46°.	Given.
$x + 58° + 46° = 180°$	Angles in a triangle add to 180°.
$x = 76°$	Solve the equation.